Resilient Rendezvous for Heterogeneous Marine-Aerial Teams Under Communication Uncertainty

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***Abstract*—This paper studies resilient rendezvous between a sustainably powered uncrewed surface vehicle (USV) and a team of aerial drones operating under intermittent communication loss. The objective is to maintain swarm agreement while tracking a moving leader whose state is only intermittently available. We formulate the problem as leader-following consensus over a switching directed graph with intermittent pinning to the USV and show why consensus alone can synchronize the drones without eliminating steady-state tracking bias. To address this issue, we augment the controller with a leader-velocity feedforward term and interpret the closed-loop behavior through a switched-system lens. In simulations with 10 drones and a nominal 40% inter-drone dropout rate, the proposed controller achieves rendezvous to a moving USV trajectory, settling in 6.65 s and reducing the final mean tracking error from 2.2730 to 0.0114 relative to a consensus-only baseline. [code][1]**

***Index Terms*—multi-agent control, leader tracking, switched systems, marine-aerial autonomy**

I. Introduction

Heterogeneous marine-aerial teams are attractive for persistent sensing, surveillance, and environmental monitoring because the platforms offer complementary mobility and endurance. A USV can provide long-duration presence, energy harvesting, and a mobile communications relay, while drones can rapidly localize transient phenomena, collect high-resolution measurements, and revisit targets of opportunity. Prior work on heterogeneous UAV–USV coordination and broader formation-control architectures demonstrates the value of cross-domain teams, but it also highlights the difficulty of maintaining coordination under mismatched sensing, actuation, and communication constraints [1], [2].

In persistent maritime sensing, synchronized rendezvous is not merely a convenience layered on top of coordination; it is the mechanism that enables data handoff, safe regrouping, recharge or servicing, and retasking between sorties. This requirement makes the problem fundamentally harder than static consensus or nominal formation keeping because the drones must agree with one another while simultaneously tracking a moving service node. The difficulty is amplified by three practical effects: the agents are heterogeneous, marine communication channels are unreliable, and the leader is moving. A controller that only enforces internal agreement can therefore succeed in synchronizing the drones while still failing the mission-level objective of physically reaching the USV.

The theoretical foundations of this problem are rooted in graph-based coordination and formation control. Early work established how information flow, Laplacian structure, and switching interaction graphs shape collective behavior in multi-agent systems [3]–[6]. Subsequent survey work has organized the literature on formation stabilization, trajectory tracking, and graph rigidity, clarifying the design tradeoffs between consensus-style controllers and formation-specific coordination laws [2], [7]. From a systems perspective, these results also connect naturally to the broader theory of switched linear systems, where stability depends on how mode transitions preserve common Lyapunov structure or sufficient dwell-time and connectivity conditions [8].

Leader-following and moving-target tracking problems extend this literature by explicitly accounting for a dynamic reference. Representative results have characterized consensus tracking with active leaders, switching topologies, and jointly connected graphs, as well as robustness to noisy measurements and intermittent observation of the leader state [9]–[12]. On the application side, recent rendezvous work for persistent tip-and-cue missions has focused on simpler host-satellite settings in which synchronization is required but the coordination structure is less heterogeneous than the marine-aerial team studied here [13]. What remains comparatively underdeveloped is the specific regime considered in this paper: multiple drones must rendezvous with a moving USV under intermittent communication loss, and the controller must resolve both topology-induced disagreement and moving-leader tracking bias.

This paper addresses that gap with the following contributions:

- We formulate synchronized rendezvous as a leader-following multi-agent tracking problem in which a USV serves as a moving reference and a drone team communicates over a switching directed graph with intermittent pinning to the leader.
- We show why a consensus-only design can synchronize the drones internally yet still leave a steady-state offset from the moving USV, motivating the addition of a leader-velocity feedforward term.

[1]Code: https://github.com/kmgovind/rob516-miniproject5

- We use a switched-system interpretation to explain why intermittent communication failure can be tolerated so long as the union of active graphs repeatedly restores sufficient connectivity for leader information to propagate.
- We demonstrate the resulting behavior in simulation for a 10-drone team tracking a moving USV under a nominal 40% dropout rate.

The remainder of the paper is organized as follows. Section II defines the system model and formal problem statement. Section III presents the control strategy and associated stability intuition. Section IV summarizes the simulation study and key observations. Section V concludes the paper and identifies open directions for a more complete study.

II. Preliminaries & Problem Statement

We consider n drones with index set $\mathcal{V} = \{1, 2, \ldots, n\}$ and a leader USV indexed by 0. The position of drone i is $x_i(t) \in \mathbb{R}^d$, and the USV position is $x_0(t) \in \mathbb{R}^d$. In the numerical study, $n = 10$ and $d = 2$, so the motion is planar. For a first-pass analysis, each drone is modeled as a single-integrator system,

$$\dot{x}_i(t) = u_i(t), \qquad i \in \mathcal{V}, \tag{1}$$

where $u_i(t) \in \mathbb{R}^d$ is the control input. The continuous-time model is discretized with the forward-Euler update

$$x_i[k+1] = x_i[k] + \Delta t\, u_i[k], \qquad \Delta t = 0.05 \text{ s}, \tag{2}$$

which corresponds to a 20 Hz position-level control update. The USV trajectory is treated as an exogenous reference generated by

$$\dot{x}_0(t) = v_0(t), \tag{3}$$

where $v_0(t)$ is bounded and piecewise continuous. In the numerical study, the USV follows the analytic circular trajectory

$$x_0(t) = \begin{bmatrix} 5\cos(0.2t) \\ 5\sin(0.2t) \end{bmatrix}, \qquad v_0(t) = \begin{bmatrix} -\sin(0.2t) \\ \cos(0.2t) \end{bmatrix}, \tag{4}$$

which has radius 5, angular rate 0.2 rad/s, and constant translational speed 1. This outer-loop model follows standard consensus-tracking abstractions used in leader-following coordination studies [9], [11].

A. Communication Model

Communication among the drones is represented by a time-varying directed graph $\mathcal{G}_{\sigma(t)} = (\mathcal{V}, \mathcal{E}_{\sigma(t)})$, where $\sigma(t)$ is a switching signal that captures link failures, packet drops, and intermittent sensing opportunities. This graph-theoretic representation is standard in consensus and switching-topology analyses [4]–[6]. If $(j, i) \in \mathcal{E}_{\sigma(t)}$, then drone i can access information from drone j at time t. The corresponding adjacency weights are denoted by $a_{ij}(t) \geq 0$, and the Laplacian matrix of the active graph is denoted by $L_{\sigma(t)}$.

In the numerical study, the nominal inter-drone graph is the complete directed graph without self-loops, and each directed edge is independently retained at every update with probability 0.6 and dropped with probability 0.4. Thus, for $i \neq j$, the implemented weights satisfy $a_{ij}(k) \in \{0, 1\}$ with i.i.d. Bernoulli switching across edges and time, while $a_{ii}(k) = 0$.

Some drones may also have direct access to the USV state. This leader-visibility pattern is described by nonnegative gains $g_i(t)$, where $g_i(t) > 0$ indicates that drone i is pinned to the USV at time t. The pinning matrix is

$$G_{\sigma(t)} = \mathrm{diag}(g_1(t), \ldots, g_n(t)). \tag{5}$$

In the numerical study, the pinning pattern is fixed rather than switching: drones 1, 2, and 3 are pinned to the USV with unit gain, while the remaining drones are unpinned. Equivalently,

$$G = \mathrm{diag}(1, 1, 1, 0, \ldots, 0). \tag{6}$$

Together, $L_{\sigma(t)}$ and G determine which information is locally available under the active communication mode.

B. Problem Statement

The rendezvous objective is to design distributed control laws $u_i(t)$ such that all drones asymptotically track the moving USV while maintaining internal consensus. Formally, we seek

$$\lim_{t\to\infty} \|x_i(t) - x_0(t)\| = 0, \qquad \forall i \in \mathcal{V}, \tag{7}$$

despite communication dropout and switching topology.

For analysis, it is useful to define the tracking error of drone i as

$$e_i(t) = x_i(t) - x_0(t). \tag{8}$$

Stacking the agent states gives $x = [x_1^T, \ldots, x_n^T]^T$ and $e = [e_1^T, \ldots, e_n^T]^T$. The goal in (7) is therefore equivalent to driving the stacked error $e(t)$ to the origin.

C. Assumptions

The analysis adopts the following assumptions to keep the formulation concise:

- The current implementation does not estimate the USV velocity. Instead, the exact analytic velocity $v_0(t)$ is supplied as a common feedforward signal to all drones, while the exact USV position $x_0(t)$ is used only by the three pinned drones.
- The switching signal is generated by i.i.d. Bernoulli packet loss on each directed inter-drone link with dropout probability 0.4 per update. The analysis still appeals to the usual joint-connectivity intuition over time [10].
- Low-level attitude and thrust dynamics are abstracted away completely: the commanded planar velocity is applied directly to the single-integrator position state, and no actuator saturation or rate limiting is modeled.

III. Methodology

Our control design combines leader-following consensus, leader-velocity feedforward, and switched-system reasoning to preserve convergence under communication loss [8]–[10]. The numerical implementation uses a single correction gain of $k = 1.5$ for both consensus and pinning.

A. Baseline Leader-Following Consensus

Ignoring leader motion for the moment, the comparison controller used in the simulation is

$$\begin{aligned} u_i(t) = & -k \sum_{j \in \mathcal{N}_i(t)} a_{ij}(t)\big(x_i(t) - x_j(t)\big) \\ & - k g_i\big(x_i(t) - x_0(t)\big), \end{aligned} \tag{9}$$

where $k = 1.5$ is the common correction gain. The first term penalizes disagreement with neighbors, while the second term pulls pinned drones toward the USV [9], [11].

In stacked form, the error dynamics induced by (9) can be written as

$$\dot{e}(t) = -k\big(L_{\sigma(t)} + G\big)e(t) - \mathbf{1}_n \otimes v_0(t), \tag{10}$$

where $\mathbf{1}_n$ is the all-ones vector and $\otimes$ denotes the Kronecker product. Equation (10) highlights a key difficulty: even if the homogeneous part is stable, a moving leader injects the disturbance term $-\mathbf{1}_n \otimes v_0(t)$, which generally creates a steady-state offset.

B. Feedforward Compensation

To cancel the common-mode motion of the USV, the implemented controller augments every agent with the exact leader velocity:

$$\begin{aligned} u_i(t) = v_0(t) - k & \sum_{j \in \mathcal{N}_i(t)} a_{ij}(t)\big(x_i(t) - x_j(t)\big) \\ & - k g_i\big(x_i(t) - x_0(t)\big), \end{aligned} \tag{11}$$

When the leader velocity is available, this common-mode feedforward offsets the drift induced by leader motion and substantially reduces or removes steady-state tracking error. In the corresponding error coordinates, the moving-reference term cancels exactly and the closed-loop dynamics become

$$\dot{e}(t) = -k\big(L_{\sigma(t)} + G\big)e(t). \tag{12}$$

This design choice is consistent with the broader leader-following tracking literature, which emphasizes the importance of propagating or estimating leader motion when the reference is not stationary [9], [12].

From an implementation perspective, (11) is attractive because the correction term still depends only on local relative-state information plus pinning. In the current implementation, no estimator is used: the exact analytic leader velocity $v_0(t)$ is injected directly as a shared feedforward input to all drones, while the leader position $x_0(t)$ enters only through the pinning term for drones with $g_i = 1$.

C. Switching-Topology Stability Intuition

The active graph may change because of packet drops or temporary loss of line-of-sight. We therefore interpret the closed-loop system as a switched linear system. The relevant resilience question is not whether every instantaneous graph is connected, but whether the network retains enough connectivity over time for information about the leader to propagate across the team [8], [10].

The key design intuition is as follows: if the union of communication graphs over each interval of length T contains a directed spanning tree rooted at a pinned agent, then disagreement modes are repeatedly damped and the swarm can continue converging despite temporary disconnections. For undirected or symmetrized interaction graphs, the second-smallest Laplacian eigenvalue λ_2 provides a useful diagnostic of instantaneous coupling strength; in the present directed setting, however, repeated joint reachability from the pinned agents over time is the more relevant condition.

This viewpoint explains the intended resilience mechanism. During dropout events, some agents may temporarily lose relative-state information from neighbors, while only the pinned agents retain direct position anchoring to the USV. Once communication links recover, leader-position information re-enters the network through the pinned agents and diffuses through the consensus term. As long as these recovery events occur often enough, the combined effect of consensus, pinning, and shared velocity feedforward still drives the team toward rendezvous.

D. Algorithmic Summary

At each control update, each drone performs the following steps:

1) Receive neighbor states from the currently available communication graph.
2) If pinned, update the local copy of the USV position; all drones receive the exact analytic USV velocity from the reference generator.
3) Compute the consensus and pinning correction, then add the common feedforward term in (11).
4) Apply the resulting position-level command with a forward-Euler update.

The current implementation uses $k = 1.5$, a forward-Euler integration step of $\Delta t = 0.05$ s (20 Hz), and an i.i.d. Bernoulli dropout model in which each directed inter-drone link is removed with probability 0.4 at every update. No actuator saturation, clipping, or rate limiting is modeled.

IV. Simulation Results

Simulations were conducted in JAX to support efficient vectorized rollout of the multi-agent dynamics. The study considered one USV following the circular trajectory $x_0(t) = [5\cos(0.2t), 5\sin(0.2t)]^T$ and a team of 10 drones initialized uniformly in $[-10, 10]^2$. Communication links were randomly dropped to emulate intermittent connectivity, with a nominal dropout rate of 40%. The purpose of the simulation was not only to show convergence, but also to distinguish between two qualitatively different behaviors: internal consensus among the drones and true rendezvous with the moving USV. The quantitative values reported below correspond to the nominal rollout shown in Figs. 1–3.

A. Simulation Setup

The USV was commanded to traverse a bounded path while the drones executed the controller in (11). We compared

TABLE I
SIMULATION SETTINGS.

Parameter	Value
Number of drones	10
USV trajectory	Radius 5 circle; angular speed 0.2 rad/s
Dropout rate	40%
Update rate	20 Hz ($\Delta t = 0.05$ s)
Correction gain	$k = 1.5$
Feedforward term	Shared leader velocity $v_0(t)$
Pinned drones	3 of 10

TABLE II
QUANTITATIVE COMPARISON WITH AND WITHOUT FEEDFORWARD.

Metric	Feedforward	Baseline
Final mean error	0.0114	2.2730
Final max error	0.0121	2.4159
Time-averaged mean error	0.4740	2.3610
Final RMSE	0.0114	2.2793
Time-averaged RMSE	0.4767	2.3682
Settling time	6.6500	N/A
Formation spread	0.0008	0.1624

two cases: a baseline leader-following consensus law without feedforward compensation and the proposed controller with the added leader-velocity term. Performance was evaluated using the mean tracking error

$$\bar{e}(t) = \frac{1}{n}\sum_{i=1}^{n} \|x_i(t) - x_0(t)\|, \tag{13}$$

as well as final maximum error, root-mean-square error (RMSE), settling time under the adopted convergence criterion, and final formation spread.

B. Qualitative Findings

The baseline controller produced a useful but incomplete behavior. The drones quickly synchronized with one another, indicating that the consensus term was strong enough to suppress internal disagreement. However, the swarm converged to a formation that lagged behind the moving USV, producing a persistent tracking error. This observation is consistent with the analysis in Section III: synchronization alone does not cancel the reference motion when the leader is moving.

After introducing the feedforward term, the qualitative behavior improved substantially. The drones still achieved internal agreement, but the collective center of the swarm now translated with the USV rather than trailing it. The quantitative comparison in Table II mirrors this visual improvement: the final mean error dropped from 2.2730 to 0.0114, the time-averaged mean error dropped from 2.3610 to 0.4740, and the final formation spread dropped from 0.1624 to 0.0008. Under the adopted convergence criterion, only the feedforward-enabled controller settled, reaching the moving-leader neighborhood in 6.6500 s.

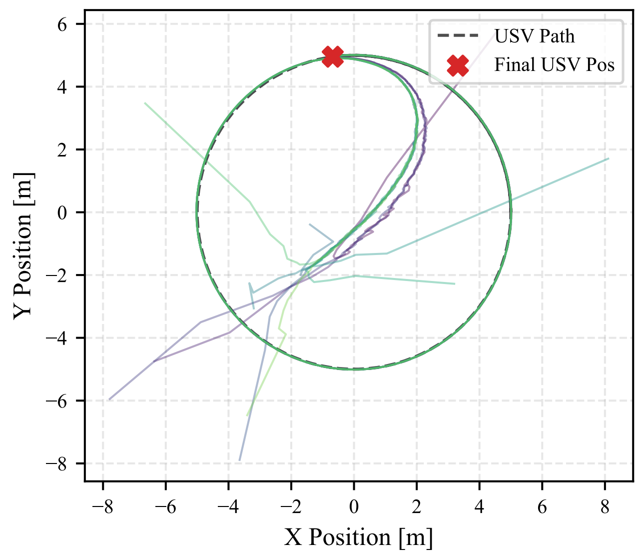

Fig. 1. Trajectories under the proposed controller with feedforward compensation. The drone team converges to the moving USV with negligible steady-state error.

C. Interpretation and Limitations

The simulation results support two main takeaways. First, intermittent communication failure does not necessarily prevent rendezvous if the switching network repeatedly regains enough connectivity for leader information to propagate. Second, explicitly compensating for leader motion is essential when the reference is not stationary. In other words, resilience to dropout and accuracy of moving-target tracking are related but distinct design requirements. The feedforward term improves not only internal agreement but also alignment of the swarm's translational motion with the leader: both the final RMSE and time-averaged RMSE remain below 0.5 with feedforward, whereas the baseline remains above 2.27 and never satisfies the settling criterion during the simulated run.

The rendezvous results for the proposed controller are shown in Fig. 1, the baseline controller response is shown in Fig. 2, and the tracking-error comparison is shown in Fig. 3.

The present study focuses on the 10-drone case and therefore does not yet characterize runtime scaling, disagreement metrics, or failure thresholds at more aggressive dropout rates and leader speeds. Those diagnostics are natural next steps for a more comprehensive evaluation.

V. CONCLUSION

This paper presented a formulation for resilient rendezvous between a mobile USV and a team of drones operating under intermittent communication loss. By combining leader-following consensus with a leader-velocity feedforward term, the proposed framework separates two key objectives: maintaining agreement within the swarm and accurately tracking a moving reference. The switched-system interpretation further

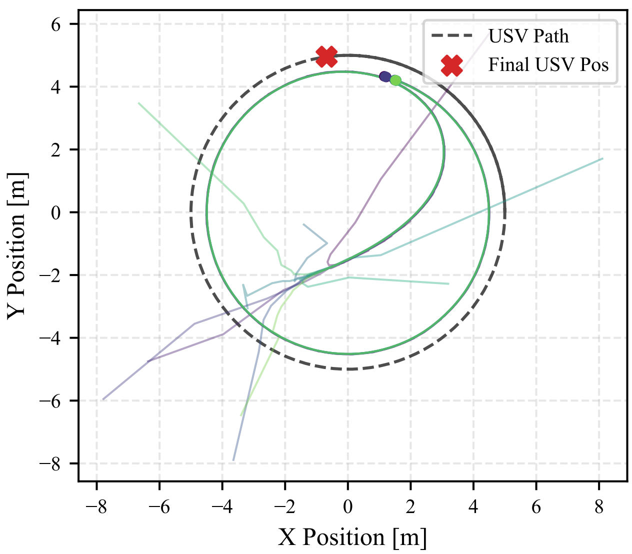

Fig. 2. Trajectories under the baseline consensus-only controller. The drones synchronize internally but retain a visible offset from the moving USV.

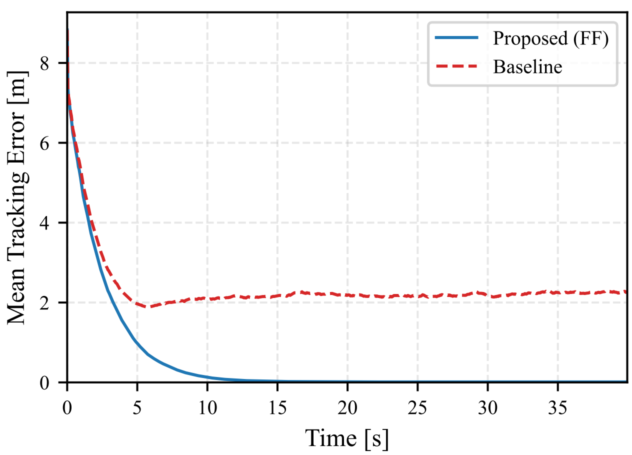

Fig. 3. Mean tracking error versus time for the proposed and baseline controllers. Feedforward compensation removes the persistent offset observed under consensus-only control.

clarifies why the team can tolerate substantial dropout so long as communication links recover often enough to preserve joint connectivity over time.

Future work will expand the analysis in three directions. First, the model can be refined to capture higher-order drone and USV dynamics, estimation error, and actuator limits. Second, the simulation study can be extended to evaluate robustness across different dropout rates, leader trajectories, and team sizes. Third, the rendezvous strategy can be integrated with energy-aware decision making, including Information Value of Energy (IVE)-style reasoning for prioritizing communication and recovery actions when drone battery reserves are low.

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